

Validation State Conditions & Applicability Module (VSCAM)

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: VALIDOS® — Validation Governance Architecture

Parent Standard: Validation State Governance Standard (VSGS)

Operational Layer: Validation State Governance Layer

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Governed Space: Validation State Conditions & Applicability

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· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Validation State Conditions & Applicability

Canonical Definition

Validation State Conditions & Applicability Module (VSCAM) defines

the governance framework for establishing, preserving, interpreting, monitoring, and enforcing the conditions, scope boundaries, restrictions, limitations, temporal validity, environmental applicability, population applicability, configuration applicability, operational applicability, and other governed constraints under which an assigned Validation State remains legitimately applicable to a Validation Object.

It establishes the operational boundary of a Validation State by defining not merely **which state the object holds**, but **where, when, how, and under what conditions that state is valid**.

Operational Role

VSCAM serves as the second internal governance space of the Validation State Governance Standard.

The preceding module establishes:

VSAEM — Is the Validation Object eligible to receive a particular

Validation State, and can that state be legitimately assigned?

VSCAM asks:

Exactly where, when, and under which continuing conditions does the assigned Validation State apply?

The governed progression is:

Assigned Validation State → Applicability Boundary Identification → Condition Mapping → Scope & Environment Mapping → Restriction & Limitation Mapping → Temporal Validity Establishment → Applicability Profile → Continuing Applicability Assessment

VSCAM does not yet govern material object change or state transition.

It establishes the conditions against which later change and monitoring can be evaluated.

Module Operational Space

VSCAM governs:

- Validation State applicability,
- state applicability boundaries,
- state scope,
- state conditions,
- continuing conditions,
- activation conditions,
- operating conditions,
- environmental conditions,
- population conditions,
- configuration conditions,
- jurisdictional conditions,
- dependency conditions,
- infrastructure conditions,
- control conditions,
- oversight conditions,
- temporal conditions,
- state restrictions,
- state limitations,
- state exclusions,
- state validity period,
- event-based validity,
- condition-based validity,
- state applicability profiles,
- applicability overlap,
- applicability fragmentation,
- applicability uncertainty,
- condition satisfaction,
- condition violation,
- applicability loss,
- and applicability traceability.

Module Function

VSCAM applies after a Validation State has been assigned or when its exact applicability must be reconstructed.

Its function is to prevent:

- a bounded Validation State being represented as universal,
- conditional validation being reported without its conditions,
- population-specific validation being generalized,
- environment-specific validation being generalized,
- configuration-specific validation being transferred across configurations,
- temporary validation being represented as permanent,
- restrictions disappearing from operational use,
- known limitations being detached from the state,
- required controls being removed without affecting validation applicability,
- state applicability continuing after a governing condition disappears,
- and the word **Validated** being interpreted without its methodological boundaries.

Validation State Applicability

Validation State Applicability describes the complete set of conditions under which a Validation State legitimately represents the Validation Object.

A state may therefore be active but applicable only within a bounded domain.

VALIDOS® establishes:

Active Validation State ≠ Universal Applicability

Applicability Profile

VSCAM may establish a formal **Validation State Applicability Profile**.

The profile may include:

- Validation Object,
- object version,
- Validation State,
- Validation Purpose,
- functional scope,
- operational scope,
- population,
- environment,
- configuration,
- jurisdiction,
- dependencies,

- required controls,
- required oversight,
- temporal validity,
- restrictions,
- exclusions,
- limitations,
- and applicable Validation Depth.

The Applicability Profile describes the operational meaning of the state.

State Scope

Every Validation State should have an identifiable scope.

State Scope may include:

- complete object,
- subsystem,
- function,
- capability,
- use case,
- operating mode,
- environment,
- population,
- jurisdiction,
- lifecycle phase,
- or another governed domain.

Scope Must Not Expand Silently

A Validation State established for:

Function A

cannot become:

Functions A, B, and C

merely because the same object performs all three.

Scope expansion requires a governed basis.

Functional Applicability

Validation may apply only to defined functions.

For example:

An AI system may be:

Validated for document classification

but not: **Validated for autonomous decision-making**

even if both functions use the same underlying model.

Operational Applicability

Validation may depend upon how the object is operated.

Examples include:

- human-supervised operation,
- autonomous operation,
- offline operation,
- real-time operation,
- low-load operation,
- high-load operation,
- normal-state operation,
- degraded-state operation,
- or emergency operation.

A state applicable in one operating mode does not automatically transfer to another.

Environmental Applicability

A Validation State may depend upon environmental conditions.

Examples include:

- temperature,
- humidity,
- altitude,
- network quality,
- lighting,
- electromagnetic conditions,
- geographic environment,
- cyber environment,
- or physical operating context.

Environmental applicability should be explicit where material.

Population Applicability

Where validation concerns human or other populations, state applicability may be population-specific.

Population dimensions may include:

- age group,
- language,
- geography,
- user type,
- professional role,
- device type,
- behavioral group,
- or another methodologically relevant characteristic.

A validated population must not silently become a universal population.

Configuration Applicability

Validation may depend upon:

- software version,
- model version,
- hardware,
- system settings,
- safety controls,
- permission structure,
- memory configuration,
- autonomy level,
- external integrations,
- or another configuration element.

Configuration is therefore part of the Validation State where it materially affects validity.

Jurisdictional Applicability

Some validation conditions may be jurisdiction-specific.

For example:

- different legal requirements,
- different technical standards,
- different populations,
- different permitted use,
- or different regulatory thresholds.

VSCAM may preserve jurisdiction as an applicability dimension without converting methodological validation into legal approval.

Dependency Applicability

A Validation Object may depend upon external components.

Examples include:

- third-party model,
- cloud infrastructure,
- external dataset,
- API,
- sensor,
- identity service,
- safety system,
- human operator,
- or another dependency.

Where validation relies materially upon that dependency, the dependency condition becomes part of state applicability.

Infrastructure Applicability

Validation may rely upon specific infrastructure.

For example:

Validated only on approved hardware configuration

or:

Validated only within Infrastructure Environment X

Migration to different infrastructure may therefore require applicability reassessment.

Control Applicability

A Validation State may depend upon a control remaining active.

Examples include:

- rate limit,
- access control,
- human approval,
- safety filter,
- isolation boundary,
- monitoring mechanism,
- fallback mechanism,
- or emergency shutdown capability.

Removing the control may affect state applicability immediately.

Oversight Applicability

A system may be validated only under a defined oversight structure.

For example:

Conditionally Validated with continuous human oversight

or:

Validated for autonomous operation up to Authority Level 2

The oversight boundary is part of validation meaning.

Activation Conditions

Some Validation States may become applicable only after defined activation conditions are met.

Examples include:

- required configuration verified,
- control activated,
- calibration completed,
- operator certification confirmed,
- dependency status verified,
- or environment checked.

State assignment and state activation may therefore be separate events.

Continuing Conditions

Some conditions must remain true throughout the lifetime of the state.

These are **Continuing Validation Conditions**.

Examples include:

- human override remains available,
 - configuration remains unchanged,
 - monitoring remains active,
 - dependency remains within approved version range,
 - operating environment remains within threshold,
 - or population remains within validated scope.
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Condition Identity

Every material condition should be sufficiently identifiable.

A Condition Record may include:

- Condition Identifier,
- condition definition,
- source,
- affected state,
- affected scope,
- activation requirement,
- continuing requirement,
- verification method,
- violation consequence,
- and reassessment trigger.

Condition Satisfaction

A condition may be classified as:

Satisfied

Conditionally Satisfied

Not Satisfied

or:

Undetermined

Where a continuing mandatory condition becomes Not Satisfied, the current state may lose applicability.

Condition Verification

Conditions should be verifiable where possible.

Verification may use:

- automated monitoring,
- system configuration check,
- evidence review,
- human confirmation,
- external source,
- sensor measurement,
- audit,
- or another governed mechanism.

Condition Verification Frequency

Some conditions need verification only at state activation.

Others may require:

- continuous verification,
- periodic verification,
- event-triggered verification,
- or verification upon material change.

The required frequency should correspond to how quickly the condition can change and how materially it affects validity.

Condition Violation

A **Validation State Condition Violation** occurs when a required condition ceases to hold.

Violation does not necessarily mean that the Validation Object itself failed validation.

It means the conditions supporting the current state are no longer fully satisfied.

Possible consequences may include:

- reduced applicability,
- Conditional State,
- Partial State,
- State Suspension,
- Revalidation Required,
- or another governed transition.

Condition Restoration

Where a violated condition is restored, the state does not necessarily reactivate automatically.

VSGS may require confirmation that:

- no material consequence occurred,
- object state remained equivalent,
- the determination remains applicable,
- and no additional reassessment is required.

Restrictions

A **Validation State Restriction** defines what the state must not be interpreted or used to cover.

Examples include:

Not validated for fully autonomous operation

Not validated outside Environment X

Not validated for Population Y

Not validated after model retraining

Restrictions define the negative boundary of applicability.

Restriction Is Not Limitation

VALIDOS® distinguishes:

Restriction — a defined boundary outside which the state must not be represented as applicable.

Limitation — a known weakness or constraint within the state or validation basis.

For example:

Restriction: Not validated for pediatric use.

Limitation: Limited long-term evidence in the validated adult population.

These have different methodological meanings.

Limitations

State limitations may include:

- limited evidence volume,
- limited duration,
- incomplete rare-event coverage,
- narrow environmental diversity,
- limited external validation,
- unresolved low-level uncertainty,
- or other known weaknesses.

Limitations do not automatically invalidate the state.

They remain part of its interpretation.

State Exclusions

An exclusion explicitly removes a defined domain from the state.

For example:

Validated except degraded-network operation.

The excluded domain may independently hold:

Validation Required

Validation Pending

Conditionally Validated

or another appropriate state.

Temporal Applicability

Validation State may have temporal boundaries.

VSCAM distinguishes several validity models.

Fixed-Time Validity

The state remains applicable until a defined date or duration.

For example:

Valid until 12 August 2027

subject to earlier change triggers.

Event-Based Validity

The state remains applicable until a defined event occurs.

Examples include:

Valid until model retraining Valid until hardware replacement

Valid until scope expansion

Valid until major software update

Condition-Based Validity

The state remains applicable only while specified conditions remain satisfied.

For example:

Valid while Human Override Control remains active.

Continuous Applicability

A state may remain active without a fixed expiration where continuing monitoring confirms that material applicability conditions remain satisfied.

Continuous applicability does not mean permanent validation.

It means continued validity is governed through ongoing condition assessment.

Hybrid Validity

A Validation State may combine validity models.

For example:

Valid for twelve months OR until model retraining, whichever occurs first, provided required monitoring remains active.

VSCAM permits such structures where operationally justified.

State Validity Start

The validity period should have an identifiable starting point.

Possible points include:

- state assignment,
 - state activation,
 - determination approval,
 - deployment activation,
 - or another governed event.
-

State Validity End

The validity end may be:

- fixed date,
 - elapsed duration,
 - triggering event,
 - condition failure,
 - object change,
 - superseding state,
 - or another defined lifecycle condition.
-

Expiration Condition

A state should specify what causes it to become **Validation Expired** where expiration is part of the validity model.

Expiration logic should be established before expiration occurs.

Grace Period

Where appropriate, a methodology may define a governed grace period.

A grace period should not silently extend the Validation State.

It should specify whether the object becomes:

- temporarily restricted,
 - Revalidation Required,
 - Validation Pending,
 - or another defined status during the grace period.
-

Applicability Intersection

A Validation State may depend upon several dimensions simultaneously.

For example:

Version 4.0

Adult Population

Environment A

Human Oversight

Jurisdiction X

The legitimate applicability domain is the intersection of those required conditions.

Applicability Fragmentation

A Validation Object may have different state applicability across different domains.

For example:

Environment A — Validated

Environment B — Conditionally Validated

Environment C — Validation Required

VSCAM preserves this fragmentation rather than forcing one universal state.

Applicability Overlap

Multiple Validation States may overlap.

Where overlap occurs, VSCAM should identify:

- which state controls,
 - whether states are compatible,
 - whether one is more specific,
 - whether one supersedes another,
 - or whether conflict exists.
-

Specificity Principle

Where two legitimate states overlap, a more specific state may govern the specific domain where the governance structure defines such precedence.

For example:

System State: Validated

but:

Degraded-Network State: Validation Suspended

The specific degraded-network state may control when that condition exists.

Applicability Conflict

An Applicability Conflict exists where two active state definitions claim incompatible conditions for the same object, scope, and time.

Such conflict should be escalated to later State Transition and Continuity governance rather than silently choosing one state.

Applicability Uncertainty

Sometimes it may be unclear whether the state applies.

Examples include:

- environment cannot be reliably classified,
 - object version cannot be confirmed,
 - dependency version is unknown,
 - required control status is uncertain,
 - population boundary is ambiguous,
 - or temporal validity cannot be reconstructed.
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VSCAM treats this as **Applicability Uncertainty**.

Applicability Uncertainty Materiality

Applicability uncertainty may be:

Immaterial

Tolerable Conditionally Tolerable

State-Limiting

or:

State-Blocking

State-blocking uncertainty may require suspension or reassessment.

Applicability Loss

Validation State Applicability Loss occurs when the assigned state can no longer legitimately represent the current object within a defined domain.

Possible causes include:

- condition failure,
- scope change,
- environment change,
- configuration change,
- dependency change,
- expiration,
- restriction breach,
- or loss of required oversight.

Applicability loss becomes an input to subsequent state-change governance.

Partial Applicability Loss

Applicability may be lost only for part of the state.

For example:

A system may remain Validated for:

Normal Operation

while losing applicability for:

Degraded Operation

VSCAM supports bounded loss rather than automatically invalidating the complete object.

Applicability Restoration

Where the conditions supporting applicability are restored, VSCAM may establish that the original applicability basis potentially exists again.

However, actual state restoration may require state-transition governance.

Therefore:

Applicability Restored ≠ State Automatically Restored

State Applicability Record

A formal record may preserve:

- Validation State Identifier,
- Validation Object,
- object version,
- state classification,
- Validation Purpose,
- scope,
- Validation Depth,
- applicable functions,
- operating modes,
- environments,
- populations,
- configurations,
- jurisdictions,
- dependencies,
- required controls,
- oversight requirements,
- activation conditions,
- continuing conditions,
- restrictions,
- exclusions,
- limitations,
- validity model,
- validity start,
- validity end,
- expiration conditions,
- applicability uncertainty,
- and sufficient traceability.

Machine-Readable Applicability

VSCAM should support machine-readable representation where useful.

A machine could theoretically query:

State: Conditionally Validated **Object Version:** 5.2 **Function:**
Decision Support **Population:** Adults **Environment:** Clinical
Environment A **Required Condition:** Human Approval Active **Restriction:**
No Autonomous Final Decision **Valid Until:** Defined Expiration Event

This enables operational systems to reason about
validation boundaries.

Runtime Applicability Checking

In mature implementations, systems may evaluate whether current
operation remains within validated applicability.

Conceptually:

Current Operational Context

vs.

Validation State Applicability Profile

If the context moves outside the profile, the system may generate:

Validation Applicability Boundary Exceeded

This is a governance signal, not automatically a system
shutdown command.

Applicability Boundary Exceeded

A boundary-exceeded signal may occur when:

- scope is exceeded,
- environment is outside validation,
- population is outside validation,
- configuration differs,
- condition fails,
- restriction is breached,
- or validity period ends.

The response belongs to later transition governance.

Automated Condition Monitoring

Machine-readable conditions may permit automated monitoring.

Examples include:

- configuration hash,
- model version,
- safety control status,
- environment threshold,
- dependency version,
- system mode,
- or expiration time.

Automated monitoring may improve lifecycle responsiveness.

AI-Assisted Applicability Governance

AI may assist with:

- extracting state conditions,
- mapping operational context,
- identifying boundary mismatch,
- detecting condition interactions,
- summarizing limitations,
- or identifying potential applicability loss.

AI should not silently redefine the validated boundary.

Human Applicability Judgment

Some applicability questions require expert interpretation.

Where human judgment is used, the basis should remain:

- attributable,
 - evidence-aware,
 - scope-aware,
 - condition-aware,
 - and traceable.
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Applicability Reassessment

Applicability may require reassessment when:

- context changes,
 - operating mode changes,
 - population changes,
 - jurisdiction changes,
 - dependency changes,
 - controls change,
 - or new limitations emerge.
-

This reassessment does not necessarily require full revalidation.

Minimum Implementation Framework

1. Identify Assigned Validation State

Establish the state whose applicability is being governed.

2. Establish Applicability Dimensions

Define relevant:

- scope,
- function,
- operating mode,
- environment,
- population,
- configuration,
- jurisdiction,
- dependencies,
- controls,
- and oversight.

3. Establish State Conditions

Identify activation and continuing conditions.

4. Establish Restrictions, Exclusions & Limitations

Preserve the boundaries inherited from the Validation Determination and state assignment.

5. Establish Temporal Validity

Define whether validity is:

- fixed-time,
- event-based,
- condition-based,
- continuous,
- or hybrid.

6. Create Validation State Applicability Profile

Represent the complete governed applicability boundary.

7. Establish Condition Verification

Define how and when material conditions are checked.

8. Detect Applicability Boundary Exceedance Identify operation outside the governed state domain.

9. Assess Applicability Uncertainty

Determine whether uncertainty affects continued applicability.

10. Record Applicability Loss Where Necessary

Preserve complete or partial loss of applicability as an input to lifecycle transition governance.

11. Preserve Traceability

Maintain:

- state,
- object,
- version,
- scope,
- conditions,
- restrictions,
- exclusions,
- limitations,
- validity model,
- environmental boundaries,
- population boundaries,
- configuration boundaries,
- dependency conditions,
- oversight conditions,
- uncertainty,
- applicability changes,
- and sufficient lifecycle continuity.

Governance Outputs

VSCAM may produce:

- Validation State Applicability Profile,
- Validation State Scope Record,
- Functional Applicability Record,
- Operational Applicability Record,
- Environmental Applicability Record,
- Population Applicability Record,
- Configuration Applicability Record,
- Jurisdictional Applicability Record,
- Dependency Applicability Record,
- Infrastructure Applicability Record,
- Control Applicability Record,
- Oversight Applicability Record,
- Validation State Condition Register,
- Continuing Validation Condition Record,
- Condition Verification Record,

- Condition Violation Record,
- State Restriction Register,
- State Exclusion Register,
- State Limitation Register,
- State Validity Model,
- State Validity Period,
- State Expiration Condition,
- Applicability Intersection Map,
- Applicability Fragmentation Map,
- Applicability Conflict Record,
- Applicability Uncertainty Assessment,
- Validation Applicability Boundary Exceeded Signal,
- Validation State Applicability Loss Record,
- Partial Applicability Loss Record,
- Applicability Restoration Record,
- and Validation State Applicability Record.

Use Case 1 — Autonomous AI

Scenario

An autonomous AI system holds:

Conditionally Validated

under the condition that human override remains continuously available.

The organization disables human override during fully autonomous operation.

Application

VSCAM detects failure of a Continuing Validation Condition.

Result

The existing state no longer remains applicable to the fully autonomous operating context.

A **Validation Applicability Boundary Exceeded** signal is generated for subsequent state-transition governance.

Use Case 2 — Healthcare AI

Scenario

A diagnostic AI system is:

Validated

for adults within a specified clinical environment.

The same model begins being used for pediatric patients.

Application

The object version has not changed.

However, Population Applicability has moved outside the validated boundary.

Result

The adult Validation State remains applicable within its original scope.

Pediatric use falls outside that state and may require a separate validation pathway.

Use Case 3 — Industrial System

Scenario An industrial control system is Validated for operation between defined temperature limits and with Safety Controller Version 7.x.

The facility upgrades to Safety Controller Version 8.0.

Application

VSCAM identifies a Dependency Applicability change.

The existing state cannot automatically be assumed applicable to the new dependency configuration.

Result

The affected state domain enters applicability reassessment rather than silently inheriting validation from the previous configuration.

Architectural Position

Validation State Conditions & Applicability is the second internal governance space of the **Validation State Governance Standard (VSGS)**.

The progression now becomes:

State Assignment & Eligibility → State Conditions & Applicability

VSAEM establishes:

Which Validation State may legitimately be assigned?

VSCAM establishes:

Exactly where, when, how, and under which continuing conditions does that state apply?

The next VSGS governance stage can then ask:

What changes in the Validation Object, its environment, dependencies, evidence, or operating reality could materially affect the continuing validity of that state, and how should those changes be detected?

The VSGS progression therefore continues:

State Assignment & Eligibility → State Conditions & Applicability → Material Change & State Monitoring → State Transition & Revalidation Triggering → State Continuity & Lifecycle Record

Foundational Principle

A Validation State has meaning only within the exact conditions and boundaries under which its validation basis remains applicable.

Canonical Closing Statement

Validation State Conditions & Applicability Module establishes the applicability-boundary layer of Validation State Governance within VALIDOS® by governing the functional, operational, environmental, population, configuration, jurisdictional, dependency, infrastructure, control, oversight, temporal, conditional, restrictive, exclusionary, and limitation boundaries of an assigned Validation State. VSCAM ensures that validation cannot silently expand beyond the conditions that justify it and creates the governed applicability profile necessary to detect when a Validation Object remains inside, approaches, or crosses the boundary of its legitimate current Validation State.

Parent Resources

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

[→ Previous module: Validation State Assignment & Eligibility Module \(VSAEM\)](#)

Related Documents

→ [Validation State Governance Standard](#)

→ [About Validation State Governance Standard](#)

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